

Periodontology 1

Classification

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Assistant Professor



Mission: To advance the science and art of health through
education, service, scholarship and social accountability

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Learning Outcomes

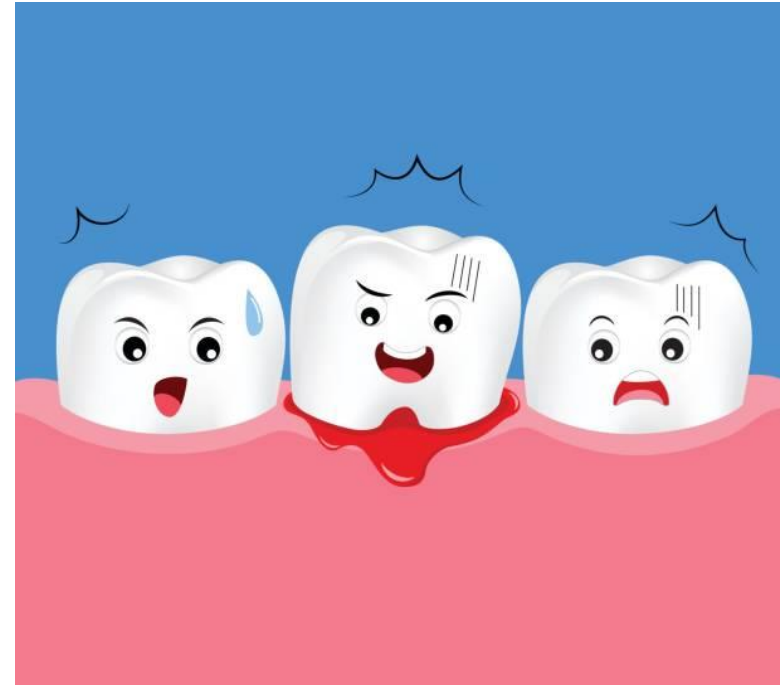
- Understand how clinical assessment influences diagnosis of disease or condition patient presents with
- Understand the AAP 1999 (Armitage) Classification and its significance
- Understand the World Workshop 2017 and 2018 findings and Classification and Diagnosis system
- Identify and classify patient's periodontal condition using the 2017/2018 system
- Relate all of the above to anatomy, BMS, patient treatment

Clinical Assessment of Periodontal Conditions

- How do we know if our patient has experienced tissue destruction/loss? Assessment
- Evaluate signs and symptoms
- Gather information, *classify and diagnose periodontal condition*, etiology, prognosis, treatment plan

Assessment of periodontal condition:

1. **Symptoms of patient**
2. **Medical and Dental History**
3. **Visual Assessment**
4. **Periodontal Examination**
5. **Radiographic Assessment**
6. **Identification of etiology**



Classification vs. Diagnosis

Classification: place in categories based on common characteristics

- Example: ICD-9/10/11 codes International Classification Diseases

Diagnosis: includes classification, information about current status of disease, and risk profile of patient

Detailed diagnosis required for best treatment plan for patient

Classification of Periodontal and Peri-Implant Diseases and Conditions

Periodontal Health, Gingival Diseases and Conditions

- Periodontal Health and Gingival Health
- Gingivitis: Dental Biofilm-Induced
- Gingival Diseases: Non-Dental Biofilm-Induced

Periodontitis

- Necrotizing Periodontal Diseases
- Periodontitis
- Periodontitis as a Manifestation of Systemic Disease

Periodontal Manifestations of Systemic Diseases and Developmental and Acquired Conditions

- Systemic Diseases or Conditions Affecting Periodontal Supporting Tissues
- Mucogingival Deformities and Conditions
- Traumatic Occlusal Forces
- Tooth- and Prosthesis-Related Factors
- Periodontal Abscesses and Endodontic-Periodontal Lesions

Peri-Implant Diseases and Conditions

- Peri-Implant Health
- Peri-Implant Mucositis
- Peri-Implantitis
- Peri-Implant Soft and Hard Tissue Deficiencies

There are two classification systems we will introduce:

1999 Armitage

The American Academy of Periodontology's (AAP) 1999 classification was based on an infection and host response model.



European Federation of Periodontology (EFP) and AAP met to define periodontal health and disease with updated information.

2017 World Workshop



World Workshop on Classification of Periodontal & Peri-implant Diseases & Conditions



The purpose of this international workshop was to use an evidence-based approach to **revise** the 1999 Classification of Periodontal Diseases and Conditions in light of new data that have emerged on the nature of periodontal diseases in the past 18 years.

1999 Classification System: Major Groups

- I. Gingival diseases
- II. Chronic periodontitis
- III. Aggressive periodontitis
- IV. Periodontitis as a manifestation of systemic disease
- V. Necrotizing periodontal diseases
- VI. Periodontal abscesses
- VII. Developmental or acquired deformities or conditions

AAP 1999 “Chronic vs. Aggressive”

- Chronic: mild, moderate, severe
Adults (typically)
Destruction=etiology
Variable microbial pattern

Slow to Moderate rate progression
Modified by systemic factors
Modified by local factors
"Generalized" or "localized"
- Aggressive “AP”:
Some Juvenile Onset “JP”, “LJP”
Destruction >etiology
A. actinomycetemcomitans
Rapid rate progression
Medically healthy
Familial aggregation
Phagocyte abnormalities
1st Molar/incisor pattern

Main changes in 2017-2018 Classification

- Periodontal health clearly defined (Lang)
- Aggressive periodontitis NOT a separate condition
- Staging and grading (medical model), progression/treatment response
- Peri-Implant diseases and conditions included and defined

A new classification scheme for periodontal and peri-implant diseases and conditions – Introduction and key changes from the 1999 classification

Jack G. Caton¹ | Gary Armitage² | Tord Berglundh³ | Iain L.C. Chapple⁴ | Søren Jepsen⁵ |
Kenneth S. Kornman⁶ | Brian L. Mealey⁷ | Panos N. Papapanou⁸ | Mariano Sanz⁹ |
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Classification

- AAP 2017



- ADA (Old-do not use!)

Type I: Gingivitis	No loss of attachment Bleeding on probing may be present
Type II: Early Periodontitis	Pocket depth or attachment loss: 3 to 4 mm Bleeding on probing may be present Localized area of gingival recession Possible Grade I furcation involvement
Type III: Moderate Periodontitis	Pocket depths or attachment loss 4 to 6 mm Bleeding on probing Grade I or II furcation involvement Class I mobility
Type IV: Advanced Periodontitis	Pocket depths or attachment loss > 6 mm Bleeding on probing Grade II or III furcation involvement Class II or III mobility
Type V: Refractory and Juvenile Periodontitis	Periodontitis not responding to conventional therapy or which recurs soon after treatment Juvenile forms of periodontitis

Staging and Grading Periodontitis



The 2017 World Workshop on the Classification of Periodontal and Peri-Implant Diseases and Conditions resulted in a new classification of periodontitis characterized by a multidimensional staging and grading system. The charts below provide an overview. Please visit perio.org/2017wwdc for the complete suite of reviews, case definition papers, and consensus reports.

PERIODONTITIS: STAGING

Staging intends to classify the severity and extent of a patient's disease based on the measurable amount of destroyed and/or damaged tissue as a result of periodontitis and to assess the specific factors that may attribute to the complexity of long-term case management.

Initial stage should be determined using clinical attachment loss (CAL). If CAL is not available, radiographic bone loss (RBL) should be used. Tooth loss due to periodontitis may modify stage definition. One or more complexity factors may shift the stage to a higher level. See perio.org/2017wwdc for additional information.

	Periodontitis	Stage I	Stage II	Stage III	Stage IV
Severity	Interdental CAL (at site of greatest loss)	1 – 2 mm	3 – 4 mm	≥5 mm	≥5 mm
	RBL	Coronal third (<15%)	Coronal third (15% - 33%)	Extending to middle third of root and beyond	Extending to middle third of root and beyond
	Tooth loss (due to periodontitis)	No tooth loss		≤4 teeth	≥5 teeth
Complexity	Local	- Max. probing depth ≤4 mm - Mostly horizontal bone loss	- Max. probing depth ≤5 mm - Mostly horizontal bone loss	In addition to Stage II complexity: - Probing depths ≥6 mm - Vertical bone loss ≥3 mm - Furcation involvement Class II or III - Moderate ridge defects	In addition to Stage III complexity: - Need for complex rehabilitation due to: - Masticatory dysfunction - Secondary occlusal trauma (tooth mobility degree ≥2) - Severe ridge defects - Bite collapse, drifting, flaring < 20 remaining teeth (10 opposing pairs)
Extent and distribution	Add to stage as descriptor	For each stage, describe extent as: - Localized (<30% of teeth involved); - Generalized; or - Molar/incisor pattern			



PERIODONTITIS: GRADING

Grading aims to indicate the rate of periodontitis progression, responsiveness to standard therapy, and potential impact on systemic health.

Clinicians should initially assume grade B disease and seek specific evidence to shift to grade A or C.

See perio.org/2017wwdc for additional information.

	Progression		Grade A: Slow rate	Grade B: Moderate rate	Grade C: Rapid rate
Primary criteria <i>Whenever available, direct evidence should be used.</i>	Direct evidence of progression	Radiographic bone loss or CAL	No loss over 5 years	<2 mm over 5 years	≥2 mm over 5 years
	Indirect evidence of progression	% bone loss / age	<0.25	0.25 to 1.0	>1.0
		Case phenotype	Heavy biofilm deposits with low levels of destruction	Destruction commensurate with biofilm deposits	Destruction exceeds expectations given biofilm deposits; specific clinical patterns suggestive of periods of rapid progression and/or early onset disease
Grade modifiers	Risk factors	Smoking	Non-smoker	<10 cigarettes/day	≥10 cigarettes/day
		Diabetes	Normoglycemic/no diagnosis of diabetes	HbA1c <7.0% in patients with diabetes	HbA1c ≥7.0% in patients with diabetes

The 2017 World Workshop on the Classification of Periodontal and Peri-Implant Diseases and Conditions was co-presented by the American Academy of Periodontology (AAP) and the European Federation of Periodontology (EFP).

Classification



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2018 Classification of Periodontal and
Peri-Implant Diseases and Conditions

Group 1 Periodontal Health, Gingival Diseases and Conditions

- A. Periodontal Health and Gingival Health
- B. Gingivitis: Dental Biofilm-Induced
- C. Gingival Diseases: Non-Dental Biofilm-Induced

A. Periodontal Health and Gingival Health

Clinical Gingival Health on an Intact Periodontium	Clinical Gingival Health on a Reduced Periodontium: Stable Periodontitis Patient	Clinical Gingival Health on a Reduced Periodontium: Non-periodontitis Patient (i.e., recession; crown lengthening, etc.)
Bleeding on Probing <10% Pocket Probing depths ≤3mm Probing Attachment Loss - No Radiological Bone Loss - No	Bleeding on Probing <10% Pocket Probing depths ≤4mm (no site ≥4mm with BOP) Probing Attachment Loss - Yes Radiological Bone Loss - Yes	Bleeding on Probing <10% Pocket Probing depths ≤3mm Probing Attachment Loss - Yes Radiological Bone Loss - Possible

Chapple et al. 2018.¹⁰



Case: Health on reduced periodontium



46 year old female, healthy

History of Phase 1 Periodontal therapy: OHI, 4 quadrants of scaling and root planing, nightguard, 3-4 month Periodontal Maintenance since then (about 5 years)

Clinical Exam: Probing depths 2-4mm, no BOP, CAL 2-3mm stable over 5 years

B. Gingivitis: Dental Biofilm-Induced



Gingivitis: Dental-Biofilm Induced

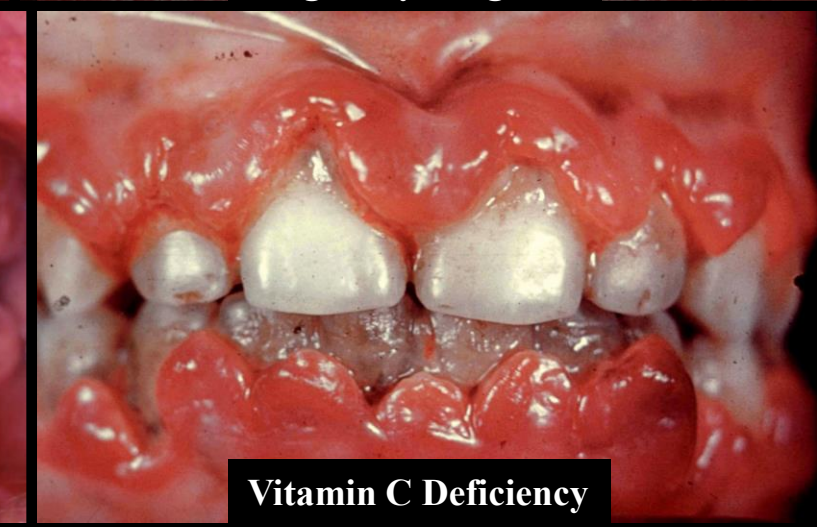
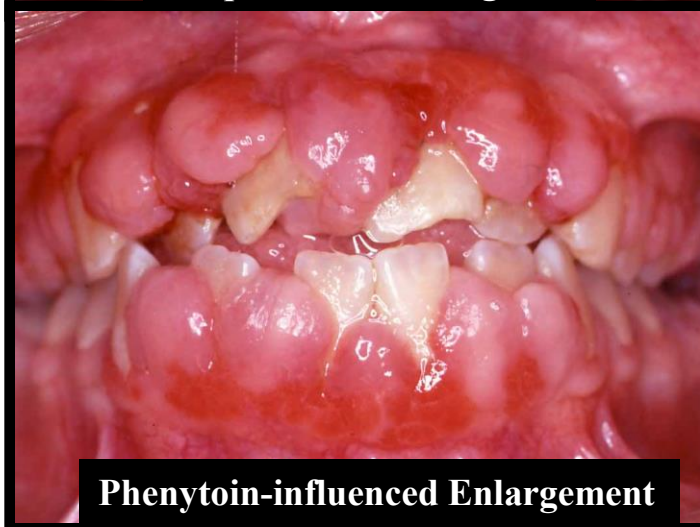
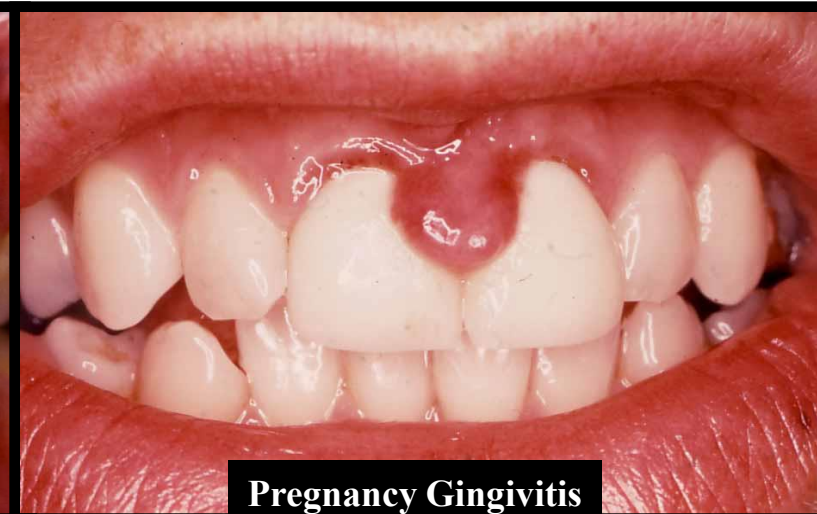
- Plaque accumulation
- Inflammatory cell infiltrate
- Weak JE (junctional epithelium) disrupted
- Increase in sulcus (probing depth)
- NO Loss Of Attachment (CAL=0)
- NO Bone loss (CEJ to crest 2mm)
- BOP: yes, Calculus: maybe
- Visual assessment:
erythematous, edematous



Classification of biofilm-induced gingivitis and modifying factors

- A. Associated with bacterial dental biofilm only
- B. Potential modifying factors of plaque-induced gingivitis
 - 1. Systemic conditions:
 - a) Sex steroid hormones (Puberty, Menstrual cycle, Pregnancy, Oral contraceptives)
 - b) Hyperglycemia
 - c) Leukemia
 - d) Smoking
 - e) Malnutrition
 - 2. Oral factors enhancing plaque accumulation
 - a) Prominent subgingival restoration margins
 - b) Hyposalivation
- C. Drug-influenced gingival enlargements

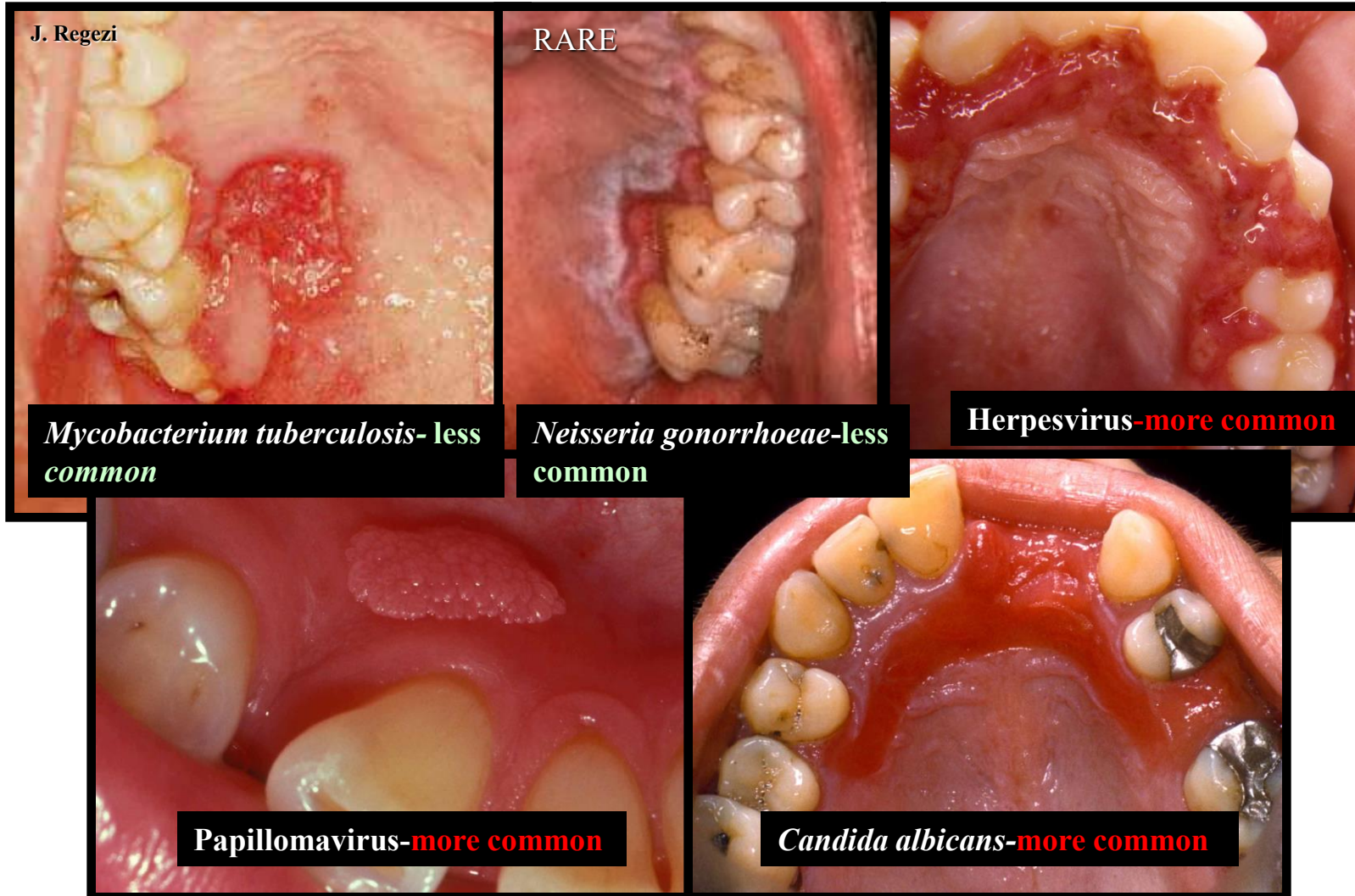
Modifiers of Biofilm-Induced Gingivitis



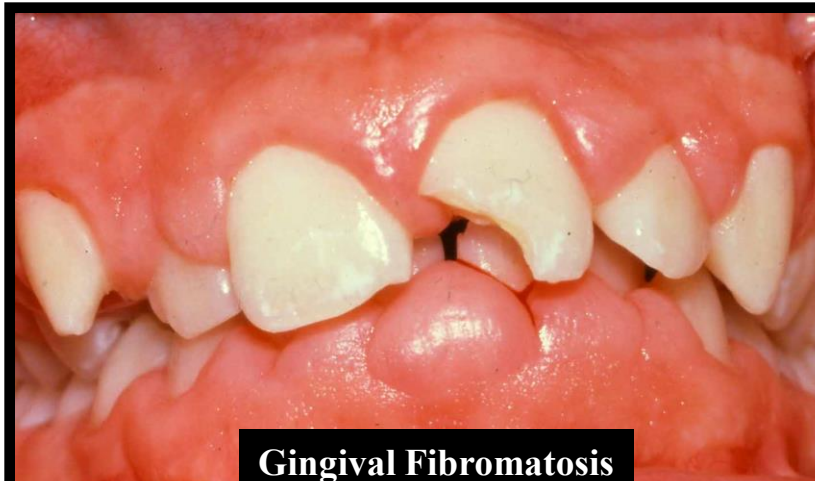
C. Non-Dental Biofilm-Induced Gingivitis

Category	Conditions		
Genetic/Developmental Disorders	Hereditary gingival fibromatosis		
Specific Infections	Bacterial Origin: <i>Neisseria gonorrhoeae</i> <i>Treponema pallidum</i> <i>Mycobacterial tuberculosis</i> - Streptococcal gingivitis Viral Origin: - Coxsackie virus (Hand-foot-and-mouth disease) - Herpes simplex I & II (Primary or recurrent) - Varicella zoster (Chicken Pox & Shingles) - Molluscum contagiosum - Human papilloma virus (squamous cell papilloma; condyloma acuminatum; verruca vulgaris; focal epithelial dysplasia)	Inflammatory & Immune Conditions	Hypersensitivity reactions: - Contact allergy - Plasma cell gingivitis - Erythema multiforme Autoimmune diseases (skin & mucous membranes) - Pemphigus vulgaris - Pemphigoid - Lichen planus - Lupus erythematosus - Systemic lupus erythematosus - Discoid lupus erythematosus Granulomatous Inflammatory Lesions (orofacial granulomatoses) - Crohn's disease - Sarcoidosis

Gingival Diseases-Non-Dental Biofilm Induced



Gingival Diseases-Non-Dental Biofilm Induced



Health and Gingivitis-literature

Case Definitions

Lang NP, Bartold MP. Gingival Health

Trombelli et al. Gingivitis

Consensus Report

Chapple et al. Gingival Health & Gingival Diseases

Literature Reviews

Murakami et al. Dental plaque-induced gingival conditions

Holmstrup et al. Non-plaque induced gingival diseases

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2018 Classification of Periodontal and
Peri-Implant Diseases and Conditions

Group 2: Periodontitis

- A. Necrotizing Periodontal Diseases
- B. Periodontitis
- C. Periodontitis as a Manifestation of Systemic Disease

A. Necrotizing Periodontal Disease

Clinical features

- Necrosis of papilla
- Ulceration
- Bleeding
- Pain
- Includes: Necrotizing Gingivitis, Necrotizing Periodontitis, Necrotizing Stomatitis (no longer “NUG” or “NUP”), NOMA= cancrum oris/gangrenous disease
- Risk factors: smoking



Classification of Necrotizing Periodontal Disease

Patient Category of Necrotizing Periodontal Disease	Patients	Predisposing Conditions	Clinical Condition
Severely & Chronically Compromised	Adult	- HIV+/AIDS with CD4 counts <200 & detectable viral load - Other severe systemic conditions that cause immunosuppression	Necrotizing Gingivitis (NG) Necrotizing Periodontitis (NP) Necrotizing Stomatitis (NS) Noma Possible progression
	Child	- Severe malnourishments - Extreme living conditions - Severe viral infections	
Temporarily or Moderately Compromised	Gingivitis Patient	- Uncontrolled factors: stress, nutrition, smoking, habits - Previous NPD (residual craters)	Generalized NG Possible progression to NP
		Local Factors: root proximity, tooth malposition	Localized NG Possible progression to NP
	Periodontitis Patient	Common Predisposing Factors	NG Infrequent Progression NP infrequent progression

Adapted from Papapanou et al. 2018.¹⁵

B. Periodontitis: Case definition

- Interdental CAL detectable at ≥ 2 non-adjacent teeth; OR
- Buccal or oral (lingual) CAL ≥ 3 mm with pocketing > 3 mm is detectable at ≥ 2 teeth.
- Absolute, unimpeachable, incontrovertible evidence of periodontitis

Periodontitis: Case definition

Causes of CAL that may not be periodontitis:

- 1) gingival recession of traumatic origin
- 2) dental caries extending in the cervical area of the tooth
- 3) CAL on the distal aspect of a second molar associated with malposition or extraction of a third molar
- 4) an endodontic lesion draining through the marginal periodontium
- 5) vertical root fracture

Main features that identify periodontitis:

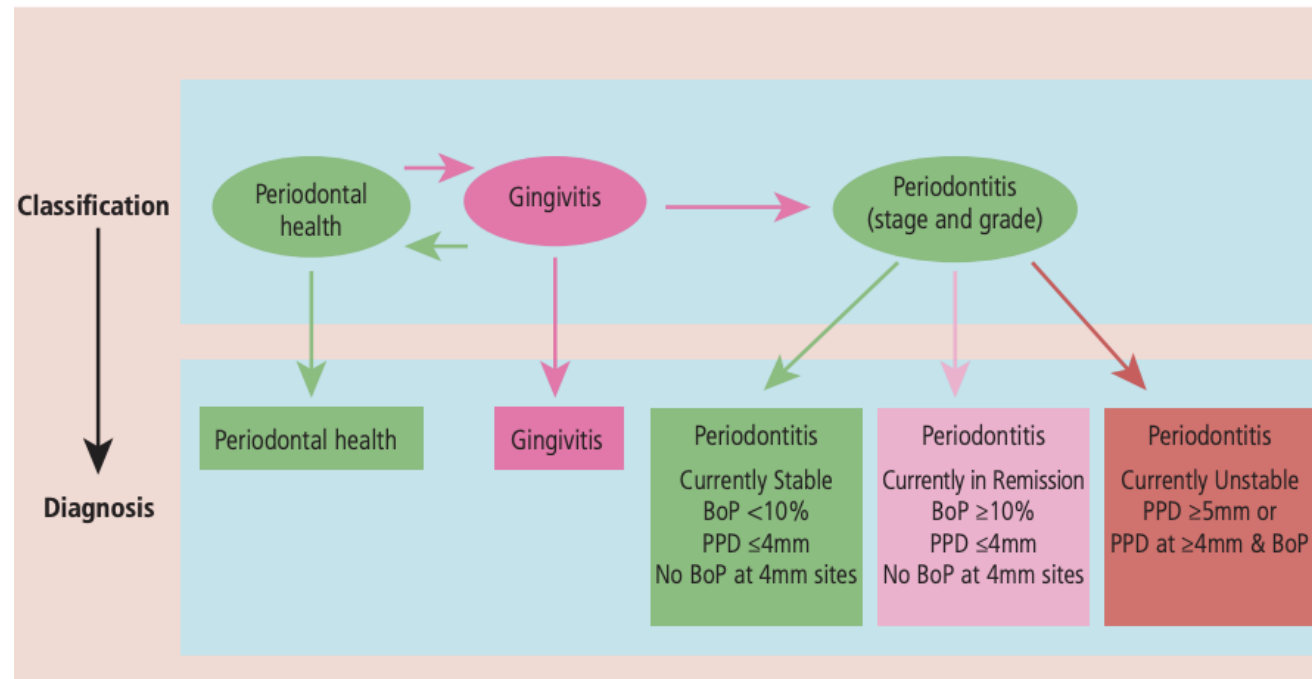
AAP World Workshop 2018

- Loss of periodontal tissue support due to inflammation
- CAL of ≥ 2 to ≥ 3 mm at ≥ 2 non-adjacent teeth
 - confirm presence of interproximal tissue loss through radiographic assessments of bone loss
- proportion of sites that bleed on probing
- proportion of teeth with probing depth over certain thresholds (commonly ≥ 4 mm and ≥ 6 mm) and of teeth with CAL of ≥ 3 mm and ≥ 5 mm

Diagnosis of Periodontitis

Once a patient has been diagnosed with periodontitis, staging and grading should be performed

Fig. 1 Possible transitions between different plaque-induced periodontal diseases. Modified from Chapple et al. 2018.¹ Classification is an important component of diagnosis, but diagnosis also includes current health/disease status, because diagnosis informs prognosis and therapeutic strategy



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Grade modifiers	Risk factors	Smoking	Non-smoker	<10 cigarettes/day	≥10 cigarettes/day
		Diabetes	Normoglycemic/no diagnosis of diabetes	HbA1c <7.0% in patients with diabetes	HbA1c ≥7.0% in patients with diabetes

C. Periodontitis as a Manifestation of Systemic Disease

- This category is comprised of uncommon systemic diseases such as Papillon-Lefevre syndrome, leukocyte adhesion deficiency.
- Alters host response to effect periodontal disease.

Periodontitis-literature

Case Definitions

- Tonetti et al. Periodontitis.
- Herrera et al. Perio abscess, necrotizing diseases, endo-perio.

Literature Reviews

- Fine et al. Aggressive periodontitis.
 - Needleman et al. Annual attachment, bone and tooth loss
 - Billings et al. Age related severity
 - RECENT PUBLICATION: Tonetti & Sanz, JCP 2019; 46:398-405.
-
- RECENT PUBLICATION: Sanz et al, J Perio 2020; 91(11):1385.

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2018 Classification of Periodontal and
Peri-Implant Diseases and Conditions

Group 3: Periodontitis as a Manifestation of Systemic Disease

- A. Systemic Diseases or Conditions Affecting Periodontal Supporting Tissues**
- B. Mucogingival Deformities and Conditions**
- C. Traumatic Occlusal Forces**
- D. Tooth- and Prosthesis-Related Factors**
- E. Periodontal Abscesses and Endodontic-Periodontal Lesions**

A. Systemic Diseases or Conditions Affecting Periodontal Supporting Tissues

Systemic conditions affecting the periodontium

Condition	Underlying pathology	Clinical features
Haematological conditions		
Neutropenia	Reduced levels of PMNLs	Severe periodontitis leading to premature loss of both deciduous and permanent dentition
Leukaemia	Bone marrow malignancy	Oral mucosal ulceration, gingival hyperplasia/hypertrophy, gingival haemorrhage, gingival pallor, periodontitis
Genetic conditions		
Down's syndrome	Defective PMNLs	Excessive plaque and calculus accumulation leading to rapid and severe periodontitis before 30 years of age; noted also in deciduous dentition
Papillon-Lefèvre syndrome	Defective PMNLs	Severe periodontitis noted in deciduous dentition, may affect permanent dentition
Chronic granulomatous disease	Defective PMNLs and macrophages	Oral ulceration but may also manifest gingivitis, periodontitis
Chediak-Higashi syndrome	Defective phagocytes	Severe childhood periodontitis
Ehlers-Danlos syndrome	Defective collagen synthesis	Childhood periodontitis
Hypophosphatasia	Defective formation/mineralization of cementum	Premature loss of deciduous teeth
Job's syndrome	Excessive IgE and histamine release by mast cells and IgE immune complex formation	Painful, bleeding gingivae, aggressive periodontitis
Histiocytosis X	Proliferation and dissemination of Langerhans cells	Gingivitis, severe periodontitis
Leucocyte adhesion deficiency syndrome	Defective PMN diapedesis	Acute inflammation, proliferation of gingival tissues, rapid bone loss, reduced wound healing
Cohen syndrome	Neutropenia	Periodontitis
Marfan's syndrome	Connective tissue disorder	Severe periodontitis
Diabetes mellitus	Increased risk of infection, reduced wound healing	Recurrent periodontal abscesses, advanced periodontitis

Adapted from [Chapple & Gilbert 2002](#). IgE: immunoglobulin E; PMNLs: polymorphonuclear leucocytes.

A



Periodontal Manifestations of Systemic Diseases and Developmental and Acquired Conditions: Consensus Report
Jepsen, Caton et al. 2018
Active link to consensus report

PERIODONTAL MANIFESTATIONS OF SYSTEMIC DISEASES AND DEVELOPMENTAL AND ACQUIRED CONDITIONS

1. Systemic diseases or conditions affecting the periodontal supporting tissues

Albandar et al. 2018 [link](#)

2. Other Periodontal Conditions

Papapanou, Sanz et al. 2018 [link](#)

Herrera et al. 2018 [link](#)

- a. Periodontal Abscesses
- b. Endodontic-Periodontal Lesions

B



3. Mucogingival deformities and conditions around teeth

Cortellini & Bissada 2018 [link](#)

- a. Gingival phenotype
- b. Gingival/soft tissue recession
- c. Lack of gingiva
- d. Decreased vestibular depth
- e. Aberrant frenum/muscle position
- f. Gingival excess
- g. Abnormal color
- h. Condition of the exposed root surface

4. Traumatic occlusal forces

Fan & Caton 2018 [link](#)

- a. Primary occlusal trauma
- b. Secondary occlusal trauma
- c. Orthodontic forces

5. Prostheses and tooth-related factors that modify or predispose to plaque-induced gingival diseases/periodontitis

Ercoli & Caton 2018 [link](#)

- a. Localized tooth-related factors
- b. Localized dental prostheses-related factors

B. Mucogingival Deformities and Conditions

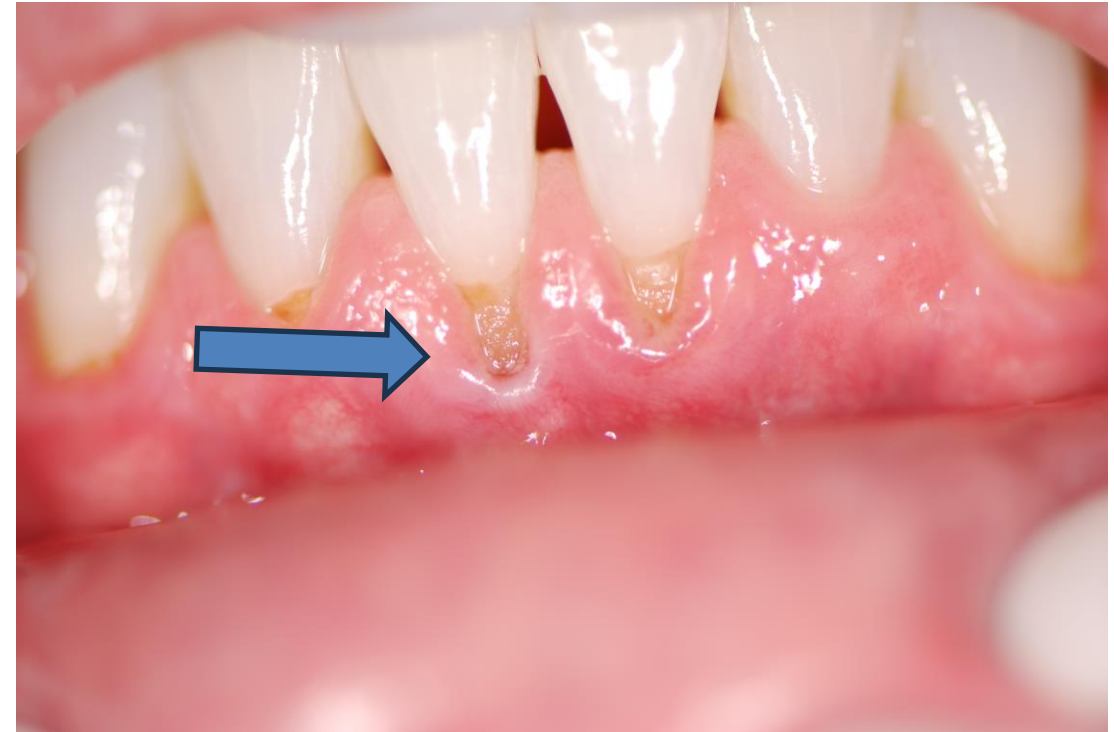
*Mucogingival deformities and conditions around teeth**

1. Periodontal biotype
 - a. thin scalloped
 - b. thick scalloped
 - c. thick flat
2. gingival/soft tissue recession
 - a. facial or lingual surfaces
 - b. interproximal (papillary)
 - c. severity of recession (Cairo RT1, 2, 3)
 - d. gingival thickness
 - e. gingival width
 - f. presence of NCCL / cervical caries
 - g. patient aesthetic concern (Smile Esthetic Index)
 - h. presence of hypersensitivity
3. lack of keratinized gingiva
4. decreased vestibular depth
5. aberrant frenum/muscle position
6. gingival excess
 - a. pseudo-pocket
 - b. inconsistent gingival margin
 - c. excessive gingival display
 - d. gingival enlargement
7. abnormal color

**Cortellini and Bissada,
2018**

Gingival Recession (GR)

- **Recession:** an apical shift of the gingival margin caused by different conditions/pathologies
- Typically associated with **clinical attachment loss**
- May apply to all surfaces
(buccal/lingual/interproximal)
- May or may not be symptomatic



Lack of Keratinized Tissue (KT)

- Attached gingiva is important to maintain gingival health in patients with suboptimal plaque control.
- Lack of keratinized tissue is considered a predisposing factor for development of gingival recessions and inflammation.
- A minimum amount of keratinized tissue is not needed to prevent attachment loss when good conditions are present.



Mucogingival Defect/Deformity (MGD)

- **Gingival phenotype (AKA Biotype)**
- The current consensus is 2 mm of **Keratinized Tissue** (KT) and 1 mm of **Attached Gingiva** (AG) are desirable around teeth to maintain periodontal health
- Less than 2mm of KT and 1mm of AG in presence of risk factors= MGD
- Other factors: restorative treatment, orthodontic treatment, frenum pull, shallow vestibular depth, tooth brushing

Mucogingival Deformities (recession)

Case Definition

Cortellini & Bissada

Consensus Report

Jepsen et al.

Discrepancy between Cortellini and Jepsen regarding Biotype and Phenotype terminology. PHENOTYPE was determined to be the recognized term.

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- Periodontal Health and Gingival Health
- Gingivitis: Dental Biofilm-Induced
- Gingival Diseases: Non-Dental Biofilm-Induced

Periodontitis

- Necrotizing Periodontal Diseases
- Periodontitis
- Periodontitis as a Manifestation of Systemic Disease

Periodontal Manifestations of Systemic Diseases and Developmental and Acquired Conditions

- Systemic Diseases or Conditions Affecting Periodontal Supporting Tissues
- Mucogingival Deformities and Conditions
- Traumatic Occlusal Forces
- Tooth- and Prosthesis-Related Factors
- Periodontal Abscesses and Endodontic-Periodontal Lesions



Peri-Implant Diseases and Conditions

- Peri-Implant Health
- Peri-Implant Mucositis
- Peri-Implantitis
- Peri-Implant Soft and Hard Tissue Deficiencies

2018 Classification of Periodontal and
Peri-Implant Diseases and Conditions

Group 4: Peri-Implant Diseases and Conditions

- A. Peri-Implant Health
- B. Peri-Implant Mucositis
- C. Peri-Implantitis
- D. Peri-Implant Soft and Hard Tissue Deficiencies

Peri-Implant Conditions

A. Peri-implant Health-absence of disease

B. Peri-implant Mucositis - is clinically similar to gingivitis in natural teeth and is characterized by clinical signs of inflammation such as bleeding on probing, possible suppuration, redness and swelling



Image Courtesy of Dr. C. Cobb.

Peri-implantitis

C. Peri-implantitis- is the next step that occurs if peri-implant mucositis is not arrested and is allowed to progress resulting in **loss of supportive bone**

D. Peri-implant Hard & Soft Tissue Deficiencies



Image Courtesy of Dr. C. Cobb.



Image Courtesy of Dr. C. Cobb.

Peri-implant Diseases and Conditions

Case Definitions

Hammerle & Tarnow. Hard and soft tissue deficiencies

Renvert et al. Peri-implant health and disease

Consensus Report

Berglundh et al. Peri-implant diseases & conditions

Literature Reviews

Araujo & Lindhe. Peri-implant health

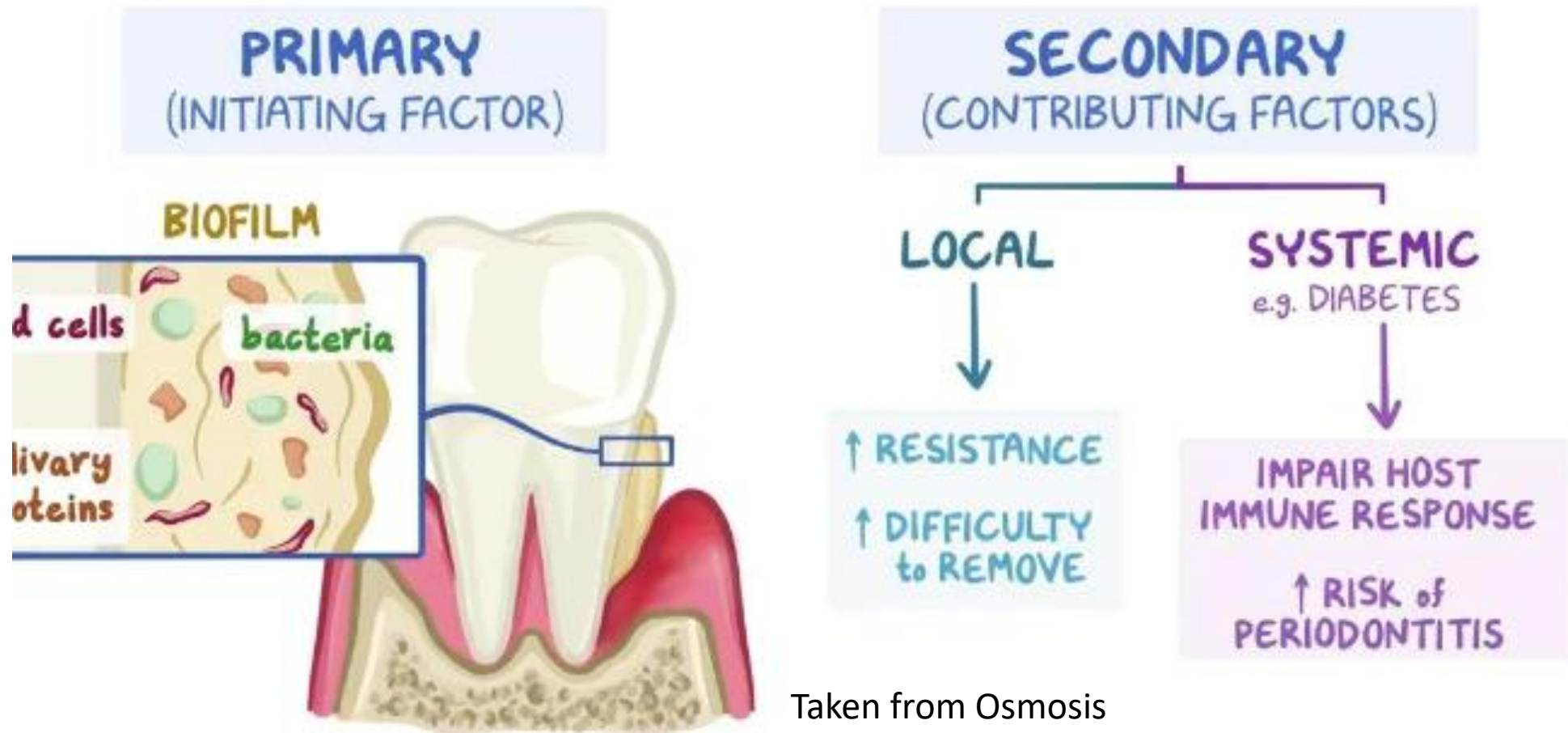
Heitz-Mayfield & Salvi. Peri-implant mucositis

Schwarz et al. Peri-implantitis

Etiology

6. Identification of Etiology

ETIOLOGIC FACTORS



Taken from Osmosis

Primary Etiology

The primary or initiating etiologic factor for periodontal disease is the dental plaque or biofilm. Without biofilms, the disease most likely will not have occurred.



Secondary Etiology

Contributing etiologic factors that enhance the ability of dental plaque to cause periodontitis. They are divided into local and systemic.

Examples of local factors:

- Iatrogenic dentistry
 - Open margins of a restoration or open contacts leading to plaque accumulation.
 - Overhanging margins making it hard to clean and creating an environment for Gram negative bacteria to grow.



Localized etiology

Specific factors can contribute to disease in an isolated area.

Examples:

- Orthodontic bracket
- Trauma
(toothbrush abrasion)
- Artifact like calculus
- Foreign body in tissue (popcorn)
- Prosthetic appliances

Examples of local etiology



Orthodontic appliances



Prosthetic appliances

From: Disease entities and pocket dentistry

Generalized Etiology

Systemic stressors activate the innate immune response of the host leading to inflammatory response and destruction of periodontal tissues.

Examples:

- Diabetes
- Stress
- Smoking
- Drugs



Image from Parker Dental

Learning Outcomes

- Understand how clinical assessment influences diagnosis of disease or condition patient presents with
- Understand the AAP 1999 (Armitage) Classification and its significance
- Understand the World Workshop 2017 and 2018 findings and Classification and Diagnosis system
- Identify and classify patient's periodontal condition using the 2017/2018 system
- Relate all of the above to anatomy, BMS, patient treatment